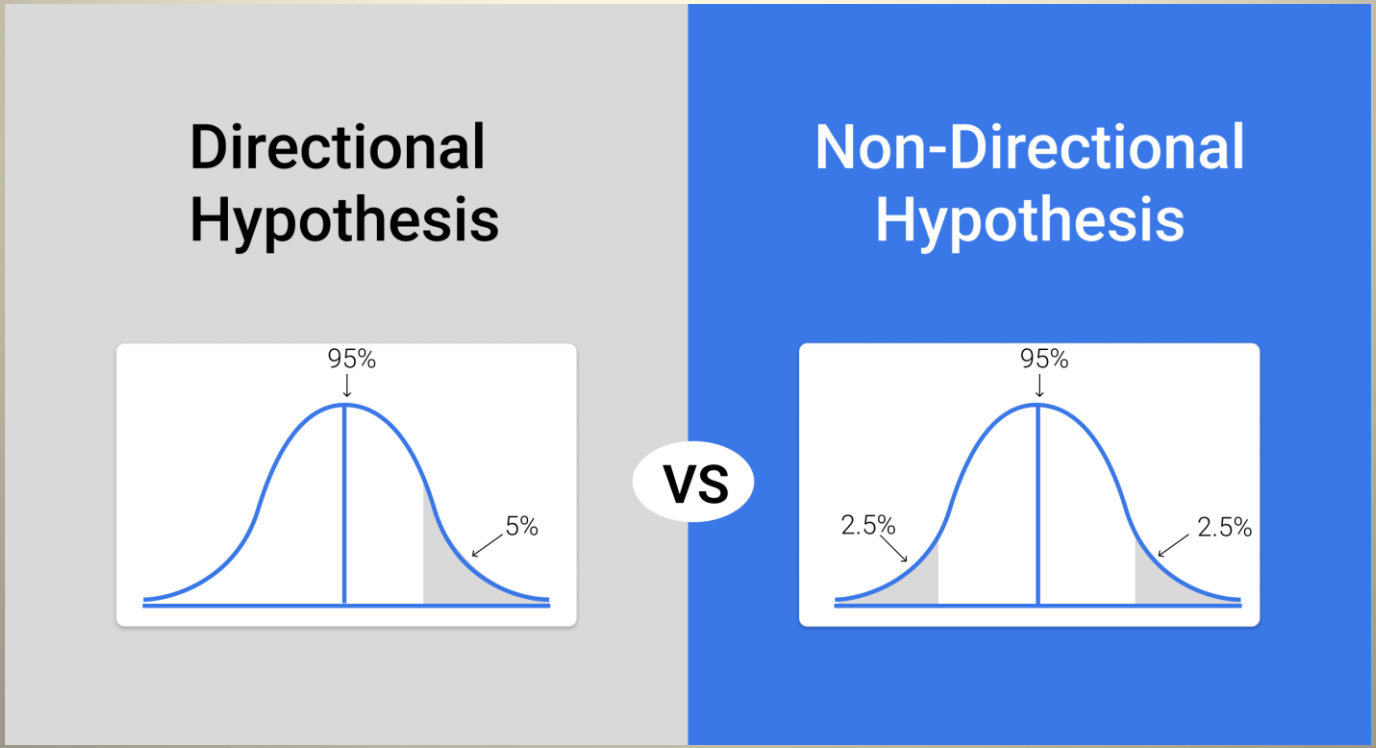
LECTURE 10

Directional Tests
&
Non-Directional Tests

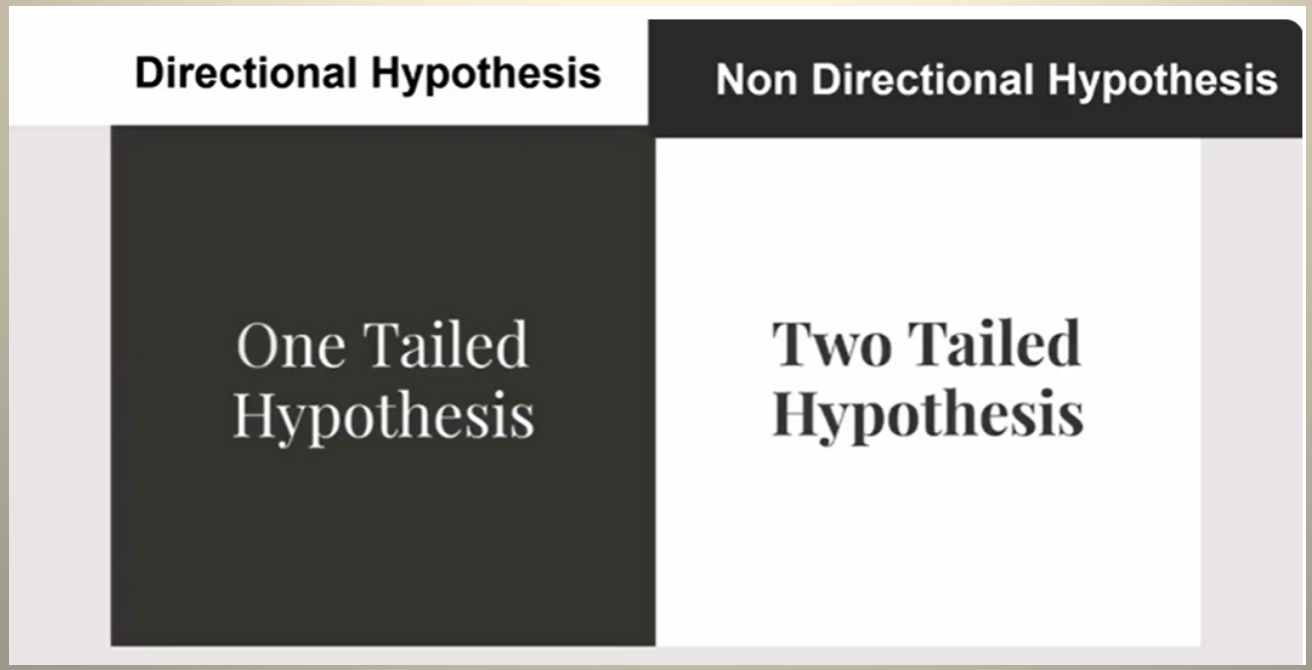


CONTENTS

- **Directional Tests**
  - ✓ One tail Test (Left Tail)
  - ✓ One tail Test (Right Tail)
- **Non-Directional Tests**
  - ✓ Two tailed test
- **Acceptance region**
- **Rejection region and Critical value Z test**

Directional Tests & Non-Directional Tests

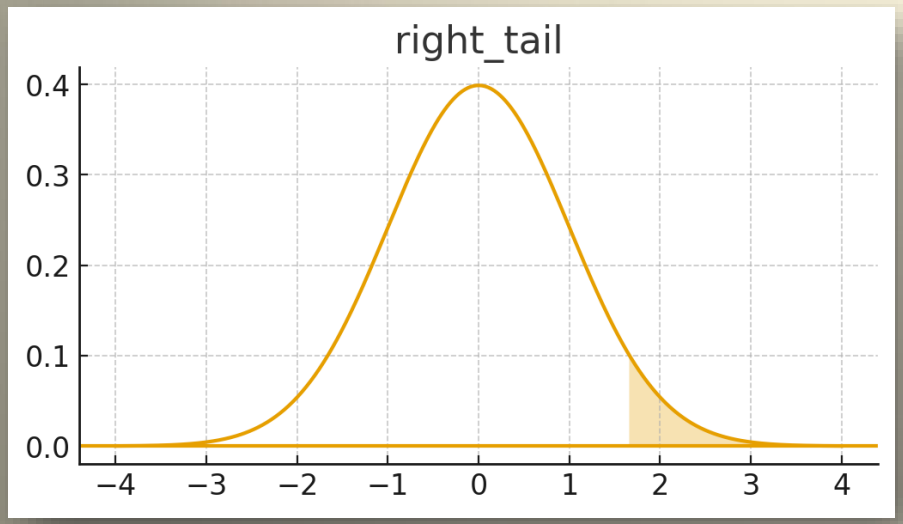
- There are two main types of hypothesis tests
  1. Directional Test
  2. Non-Directional Test

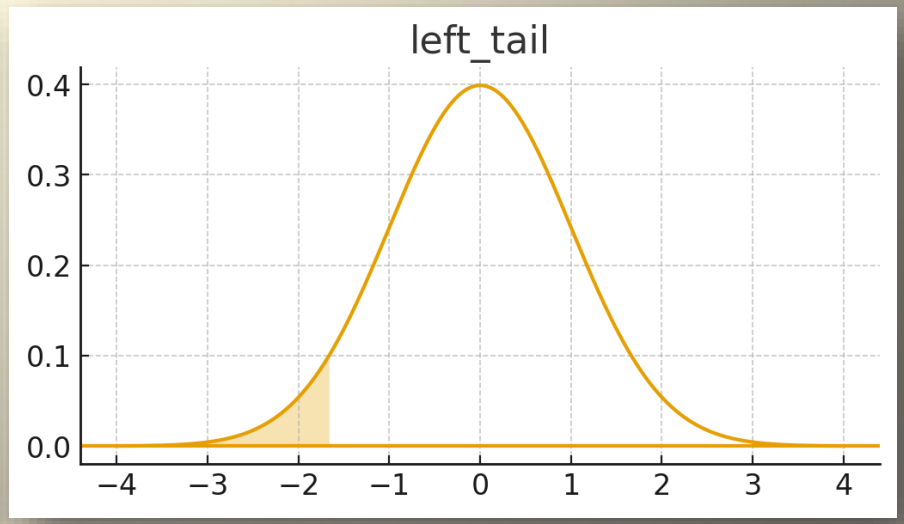


Directional Tests & Non-Directional Tests

1. Directional Test

- ✓ It is also called One-Tailed Test.
- ✓ One-tailed test, predicts the specific nature or direction of an effect.
- ✓ So, it is unidirectional (Left or Right).
  - **i. One-tailed Test (Right Tail)**
  - **ii. One-tailed Test (Left Tail)**



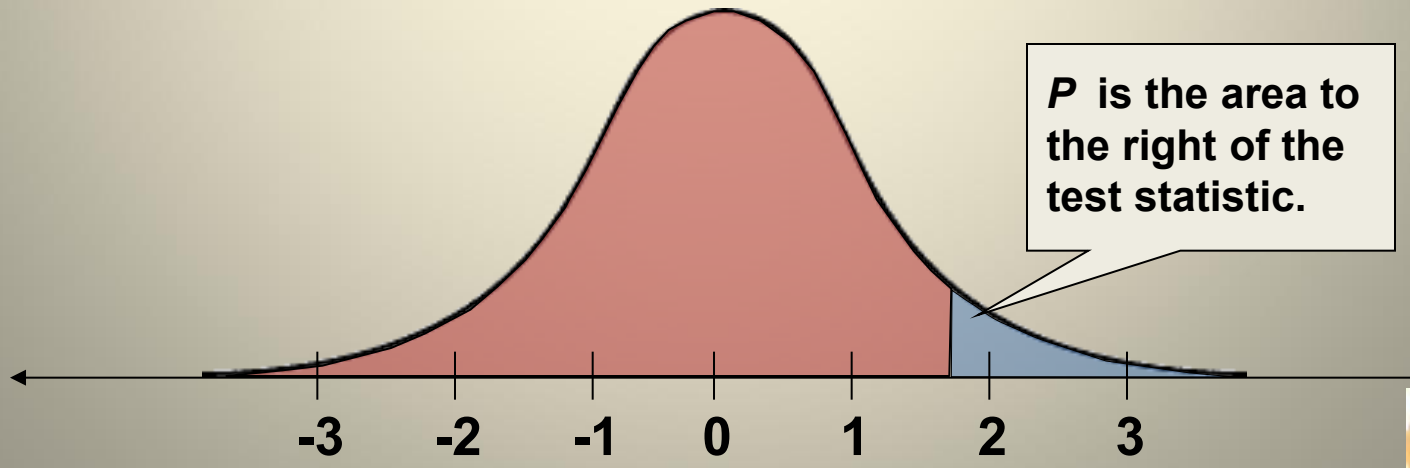


Directional Tests & Non-Directional Tests

1. Directional Test

i. One-tailed Test (Right Tail)

- If the alternative hypothesis (H_a) contains the inequality symbol of greater-than (>), the hypothesis test is a right-tailed test.
- "Exercise **increases** weight loss"
- **Example:**
  - ✓ Null hypothesis (H_0): $\mu = k$
  - ✓ Alternative hypothesis (H_a): $\mu > k$

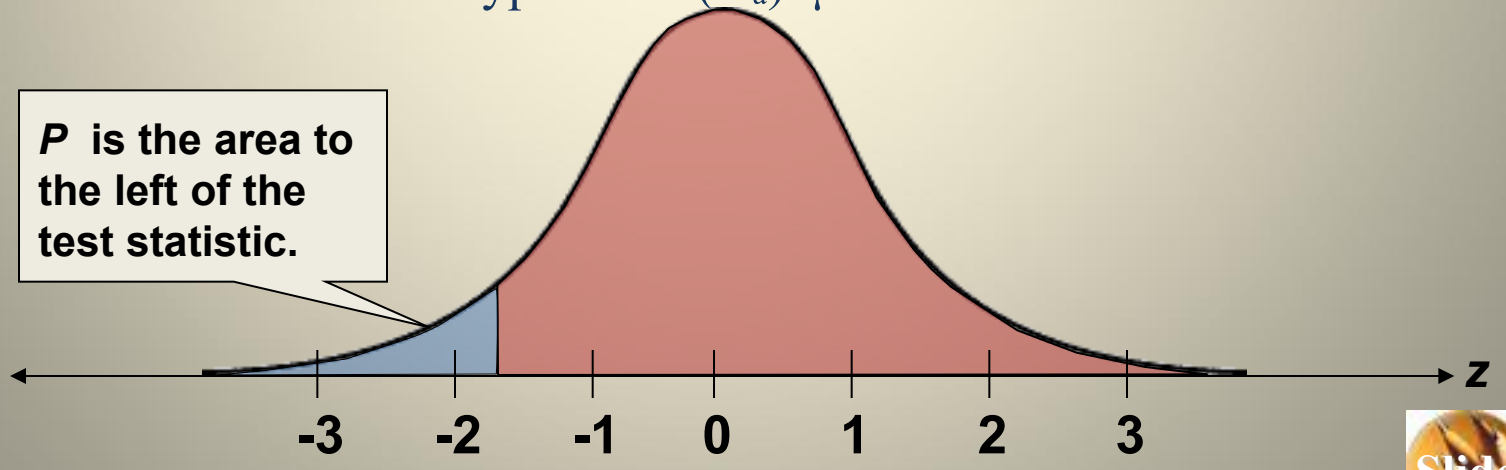


Directional Tests & Non-Directional Tests

1. Directional Test

ii. One-tailed Test (Left Tail)

- If the alternative hypothesis (H_0) contains the inequality symbol of less-than (<), the hypothesis test is a left-tailed test.
- "Exercise decrease weight loss"
- **Example:**
  - ✓ Null hypothesis (H_0): $\mu = k$
  - ✓ Alternative hypothesis (H_a): $\mu < k$



Directional Tests & Non-Directional Tests

1. Directional Test

One-tailed Test (Right Tail) Vs One-tailed Test (Left Tail)

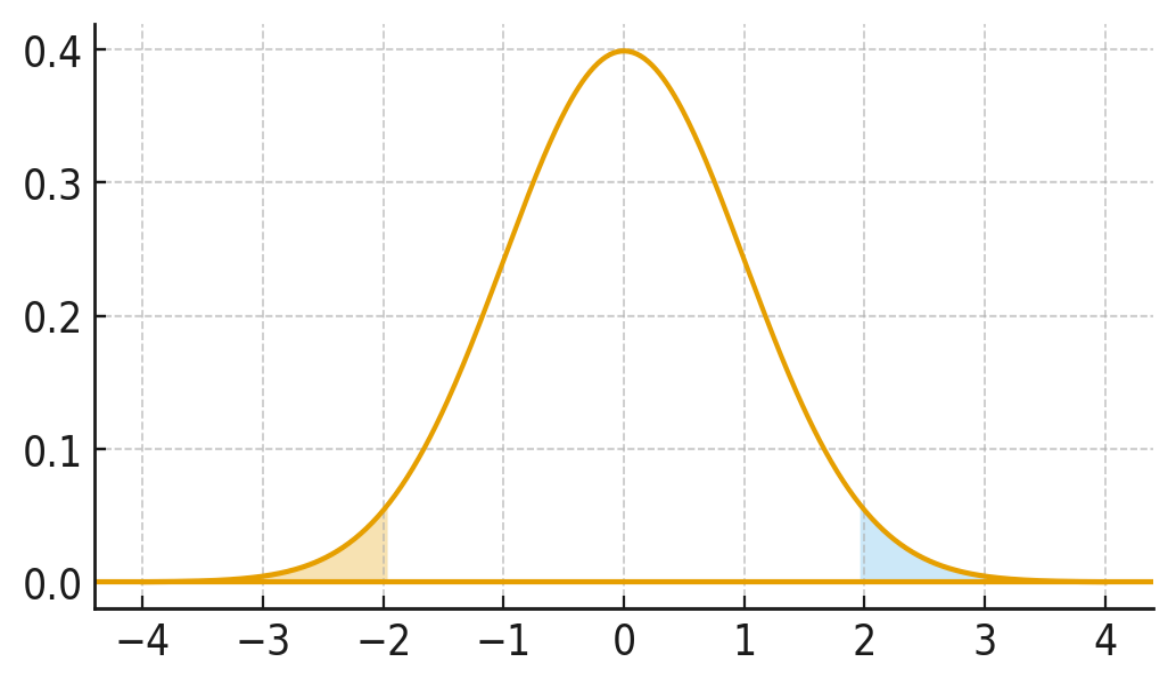
Comparison Table

Type of Test	Inequality Symbol in H_a	Meaning / Interpretation	Example Statement	Hypotheses
Right-Tailed Test	**Greater-than (>)**	Tests if the mean is **greater** than k	"Exercise increases weight loss"	H_0: $\mu = k$ H_a: $\mu > k$
Left-Tailed Test	**Less-than (<)**	Tests if the mean is **less** than k	"Exercise decreases weight loss"	H_0: $\mu = k$ H_a: $\mu < k$

Directional Tests & Non-Directional Tests

2. Non-Directional Test

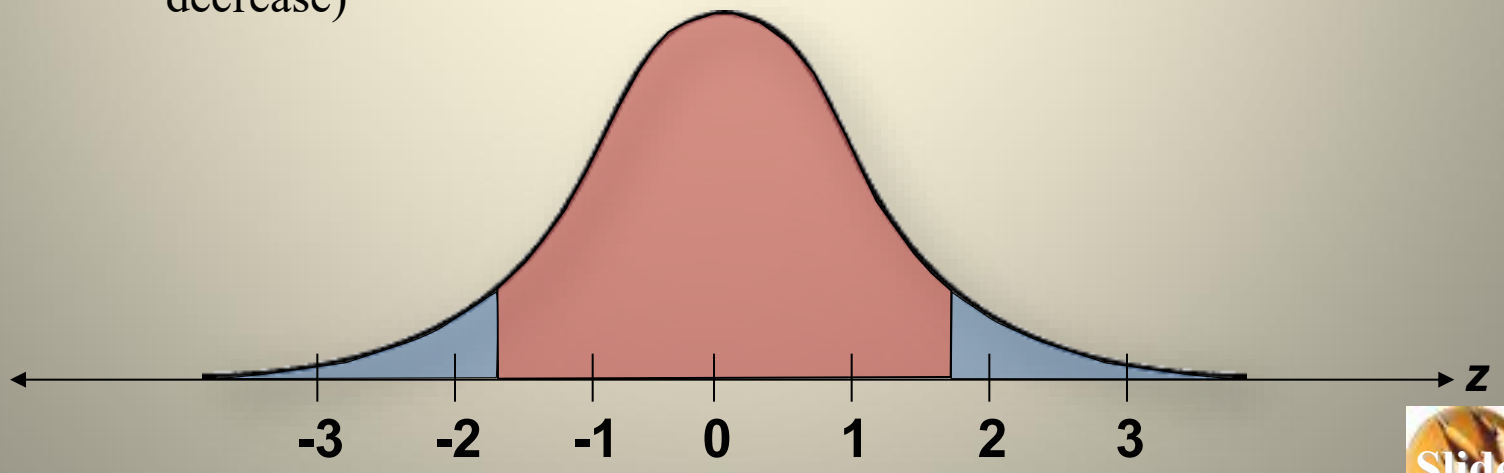
- It is also called Two-Tailed Test.
- Two-tailed test, doesn't predict the specific nature or direction of an effect.
- So, it is bidirectional (Left and Right).



Directional Tests & Non-Directional Tests

2. Non-Directional Test

- If the alternative hypothesis (H_0) contains the not-equal-to symbol ($\neq$), the hypothesis test is a two-tailed test.
- **Example:**
  - ✓ Exercise affect weight loss.
    - ❖ Null hypothesis (H_0): $\mu = k$ (no difference)
    - ❖ Alternative hypothesis (H_a): $\mu \neq k$ (there is a difference, either increase or decrease)



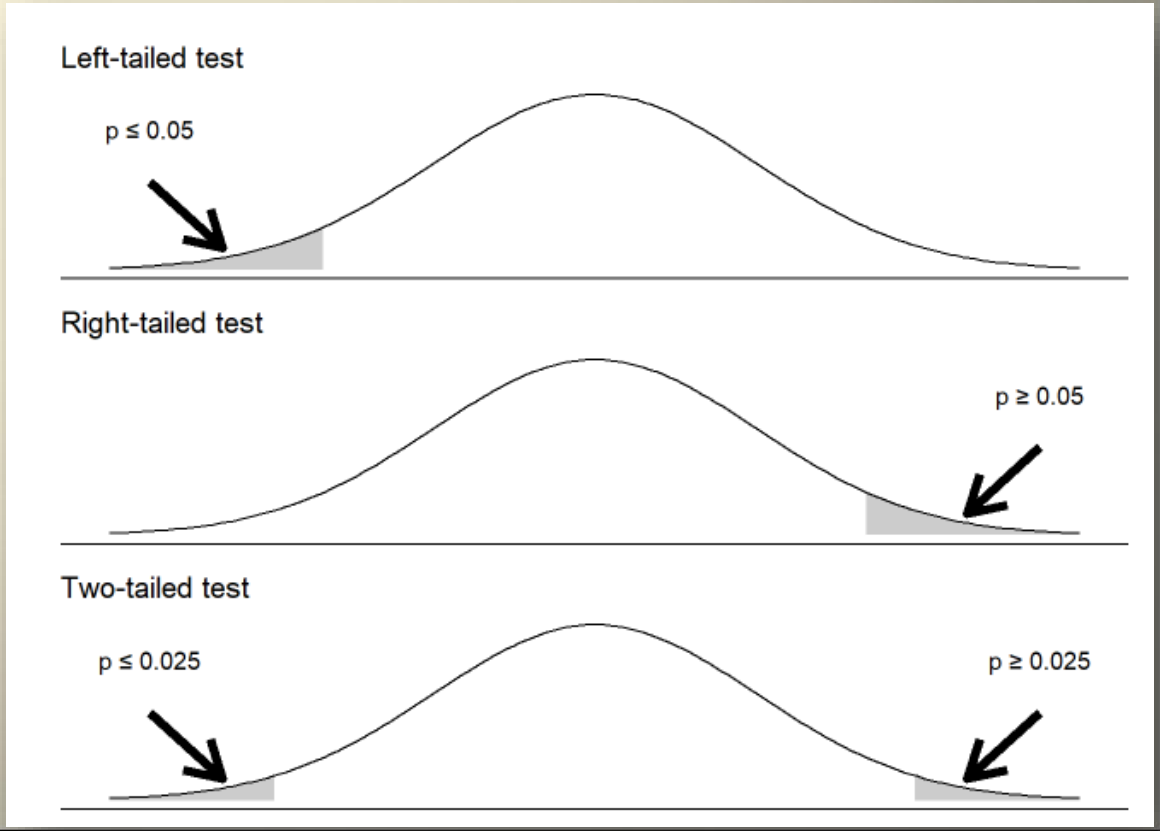
Directional Tests & Non-Directional Tests

Comparison Table

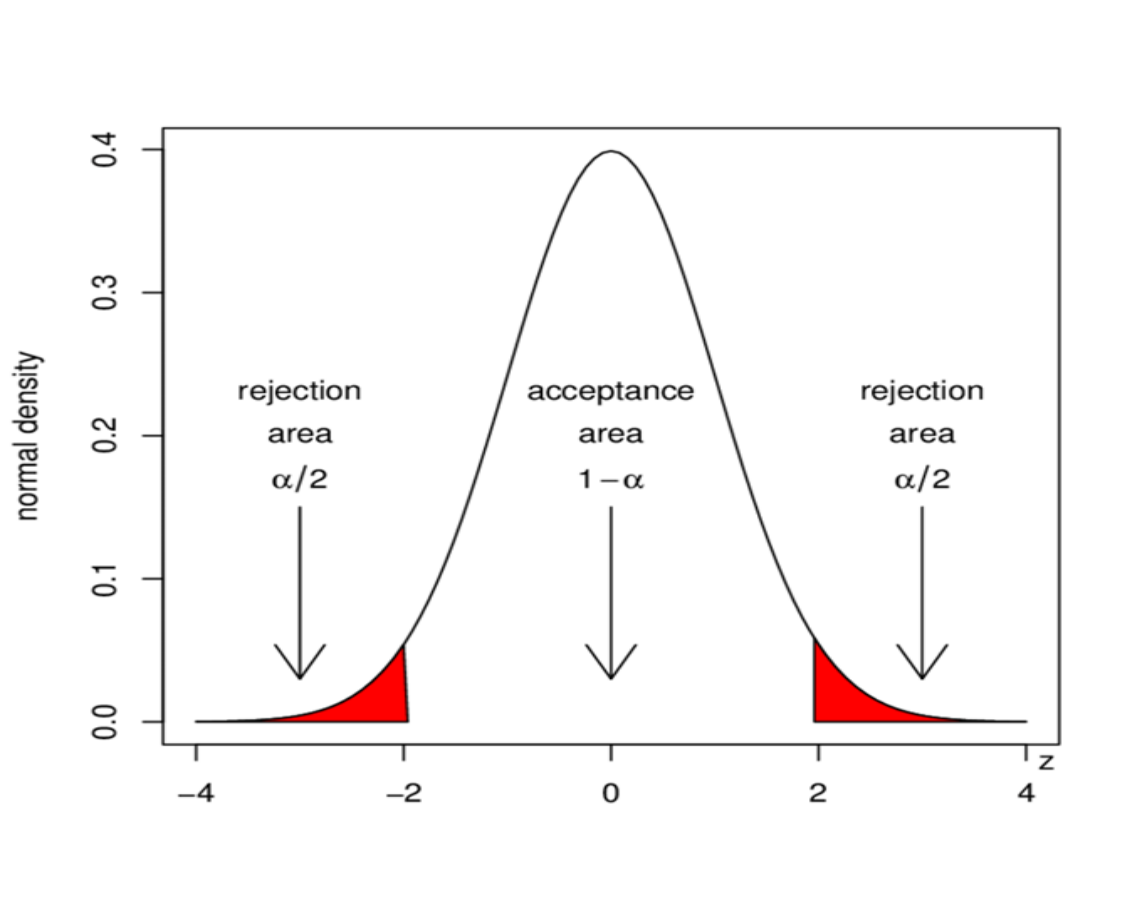
Feature	Directional Test (One-Tailed)	Non-Directional Test (Two-Tailed)
Direction	Unidirectional (Left or Right)	Bidirectional (Both Left and Right)
Specificity	Specific about the outcome (positive or negative)	Not specific about the outcome
Symbol	Uses "<" or ">"	Uses "≠"
Tails	One-tailed	Two-tailed
Definition	Hypothesis predicts the *specific nature* or *direction* of the effect	Hypothesis predicts an *effect or difference* but not the direction
When to Use	When prior research or theory strongly suggests an *increase or decrease* in a specific direction	When no clear prior evidence exists; only that a difference/effect is expected
Example	"Exercise increases weight loss" "Girls perform better than boys"	"Exercise affects weight loss" "There is a difference in performance between girls and boys"

Directional Tests & Non-Directional Tests

- Thus, there are a total of three sub-types of hypothesis tests.
  - ✓ Left-tailed Test
  - ✓ Right- tailed Test
  - ✓ Two-tailed Test



- **Acceptance region**
- **Rejection region and Critical value Z test**



Rejection Region, Acceptance Region & Critical Value (Z-Test)

Rejection Region

- The rejection region is the range of values where the null hypothesis (H_0) is unlikely to be true.
- If the test statistic falls in this region, we reject H_0.
- It is basically the area of the sampling distribution in the extreme tails.
- Rejection region: extreme values → reject H_0

Acceptance Region

- The acceptance region is the range of values where the null hypothesis (H_0) is likely to be true.
- If the test statistic falls in this region, we fail to reject H_0.
- It is basically the area of the sampling distribution not in the extreme tails.
- Acceptance region: central values → fail to reject H_0

Rejection Region, Acceptance Region & Critical Value (Z-Test)

Critical Value (Z-Test)

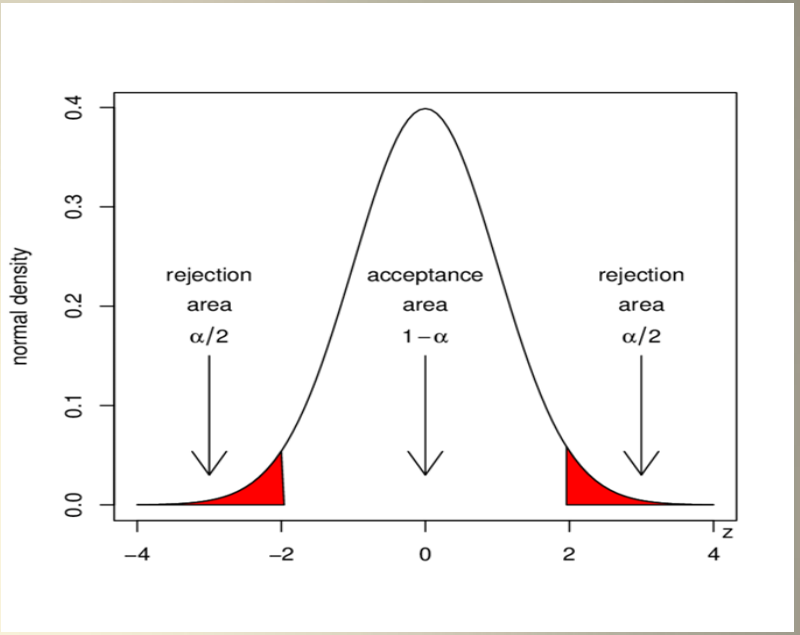
- The cutoff point that separates the rejection region from the acceptance region.
- Determined using significance level (α) and test type (left, right, two-tailed).

Example:

Find the critical value and rejection region for a **right tailed test** with **$\alpha = 0.01$**.

Rejection Region, Acceptance Region & Critical Value (Z-Test)

Finding Critical Values:



Step 1:

- ✓ Decide the significance level (α)

Step 2:

- ➢ Decide the type of test
  - ✓ Left-tailed → H_a: $\mu < k$
  - ✓ Right-tailed → H_a: $\mu > k$
  - ✓ Two-tailed → H_a: $\neq \mu \neq k$

Step 3:

- ✓ Find the critical *z* value(s) (z_0)
  - •Left-tailed test: Find z_0 such that the area to the left = α (equivalently, area to the right = 1− α)
  - •Right-tailed test: Find z_0 such that the area to the right = α (equivalently, area to the left = 1−α)
  - •Two-tailed test: Find z_0 values such that α/2 is in each tail

Step 4: Draw a normal curve

- ✓ Draw vertical line(s) at the critical value(s)
- ✓ Shade the rejection region(s)

Rejection Region, Acceptance Region & Critical Value (Z-Test)

Finding Critical Values:

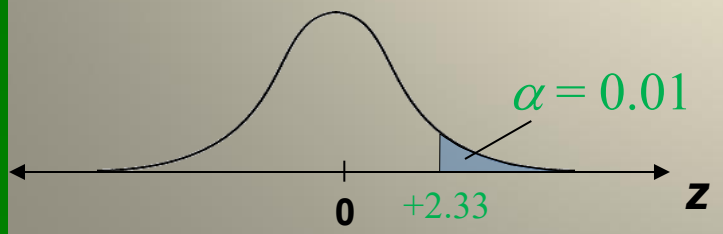
Example: Find the critical value and rejection region for a **right tailed test** with $\alpha = 0.01$.

- **Step 1: Decide the significance level (α)**
  $\alpha = 0.01$

- **Step 2: Decide the type of test**
  Right-tailed test

- **Step 3: Find the critical z-value (z_0)**
  Right-tailed test → area to the right of $z_0 = \alpha = 0.01$
  Equivalently, area to the left of $z_0 = 1 - \alpha = 1 - 0.01 = 0.99$
  From standard z-tables: area 0.9900 corresponds approximately to $z_0 \approx +2.33$

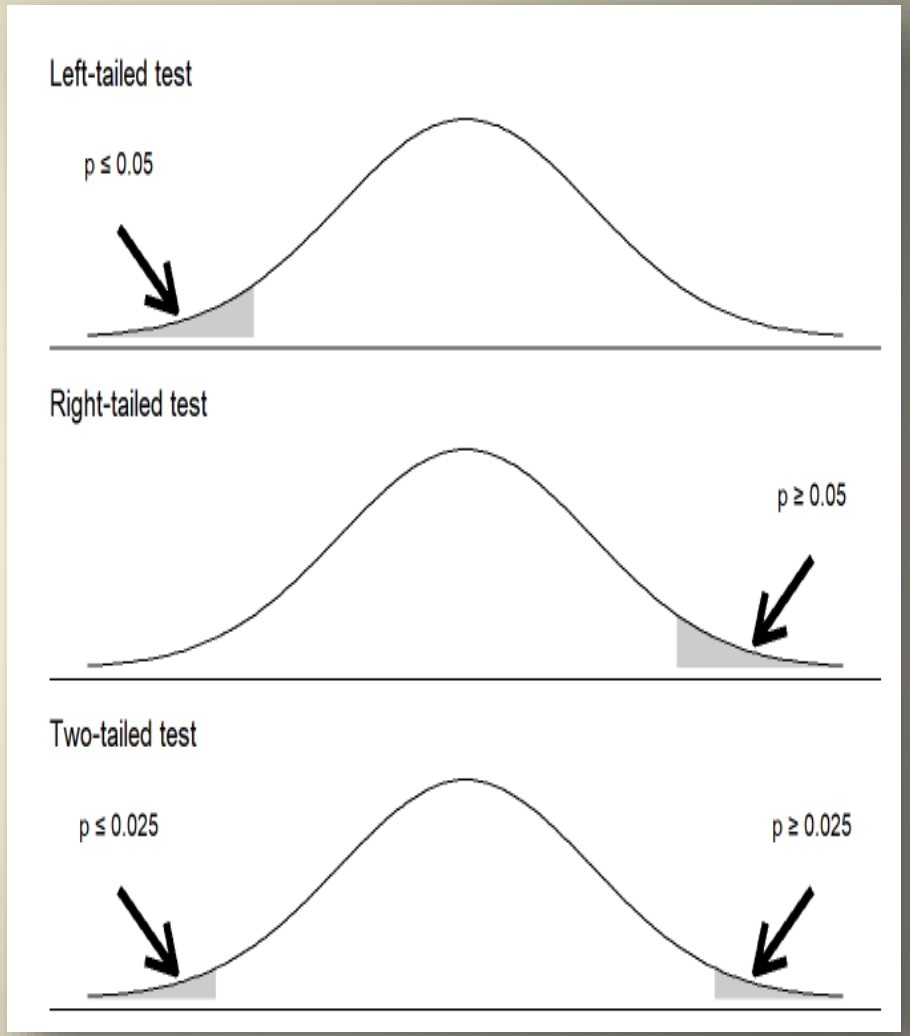
- **Step 4:** Draw a normal curve



The rejection region is to the right of $z_0 = 2.33$

Important Critical Z-values

- **Left-Tailed Tests**
  - ✓ $\alpha = 0.05 \rightarrow z = -1.645$
  - ✓ $\alpha = 0.01 \rightarrow z = -2.33$
  - ✓ $\alpha = 0.001 \rightarrow z = -3.09$
- **Right-Tailed Tests**
  - ✓ $\alpha = 0.05 \rightarrow z = +1.645$
  - ✓ $\alpha = 0.01 \rightarrow z = +2.33$
  - ✓ $\alpha = 0.001 \rightarrow z = +3.09$
- **Two-Tailed Tests**
  - ✓ $\alpha = 0.05 \rightarrow z = \pm 1.96$
  - ✓ $\alpha = 0.01 \rightarrow z = \pm 2.575$
  - ✓ $\alpha = 0.001 \rightarrow z = \pm 3.09$



📌 **You don't calculate these Z values by hand, they come directly from the Z-table or calculator.**


thankYou